Evaluating and Improving Similarity-Aware Train-Test Splitting for Wildlife Re-Identification

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Abstract. Re-identifying an individual animal from an image is a task that remains difficult even for people, since two photographs of the same animal can look different depending on when, where, and under what lighting they were taken. The WildlifeReID-10k dataset was introduced to push research beyond simple species classification and towards the harder problem of individual re-identification, and its main contribution is a splitting methodology, a time-aware split where timestamps are available, and a similarity-aware split otherwise, that groups near-duplicate images together using single-linkage clustering on DINOv2 features and keeps whole groups on one side of the training/testing boundary. In this proposal we examine this splitting methodology in detail, including a documented revision to the algorithm between the dataset's preprint and its peer-reviewed camera-ready version, and identify several open questions worth investigating: how much the similarity-aware split actually recovers of the leakage-correction effect that the time-aware split achieves on datasets where both can be compared directly, whether the clustering step is vulnerable to the well-known chaining effect of single-linkage clustering, and whether the manually-chosen, per-dataset similarity threshold at its core can be replaced with something more principled and reproducible. We lay out both these problems and a concrete plan for investigating them.

Keywords: Animal re-identification · Wildlife re-identification · WildlifeReID-10k · Clustering · Single-linkage clustering · Self-supervised learning · DINOv2.

1 Introduction

Re-identification is the task of deciding whether two images show the same individual. For animals, this is a task that is difficult even for a person doing it by hand: telling apart two photographs of lions, for instance, is not simply a matter of skill, since the two photographs may have been taken at different times, in different locations, and under different lighting conditions, and any of these differences alone can make the same individual look like a different one. This is a fundamentally different problem from the one that datasets, such as ImageNet were built to address. ImageNet and similar datasets were built around classification and object detection, where the goal is to say *what* is in an image (a lion,

a bird, a turtle). Re-identification instead asks *which* individual is in the image, so it requires a model to pick up on fine, individual specific detail rather than species level or category level detail.

Beyond the difficulty of the task itself, being able to re-identify individual animals matters because it is what makes long-term animal tracking possible. If a specific individual can be recognised again in a later photograph, weeks or months after it was first seen, then it becomes possible to follow what has happened to that animal over time, which would otherwise require much more labour-intensive manual tracking [7]. A concrete example of why this is useful: a conservationist who treats a sick or injured wild animal with medication wants to know afterwards whether the treatment actually worked, which means being able to find and recognise that same individual again in later camera-trap or survey images, rather than only being able to say that "some animal of this species" was seen again. The same reasoning applies to monitoring population size, tracking migration, and studying how individual animals respond to environmental change, all of which depend on reliably matching individuals across images taken at different times, and this is precisely why re-identification, rather than only classification, is the task that needs to be solved well.

The WildlifeReID-10k dataset [2] was introduced specifically to support this individual level task. In its peer-reviewed camera-ready form, it aggregates 37 existing wildlife re-identification datasets into a single, cleaned collection of 140,488 images of 10,772 individuals across around 33 species, at a total size of only 26GB.[1] Beyond aggregation, its central methodological contribution is a way of splitting the data into training and testing sets that tries to avoid an easy trap: because many images come from the same encounter, the same camera session, or even the same video, a naive random split can place near-duplicate images on both sides of the split, which makes the re-identification task look artificially easy and inflates reported performance. WildlifeReID-10k's proposed fix has three steps. First, around 5% of identities in each dataset are set aside entirely for open-set testing. Second, for the remaining identities, a *time-aware split* is used wherever reliable timestamps are available: the oldest roughly 85% of an individual's images go to training and the rest to testing, guaranteeing that images from the same encounter cannot end up on both sides. Third, since only

[1] We note this explicitly because the dataset has a documented revision history that matters for this proposal: an earlier, June 2024 arXiv preprint of the same paper [1] described a collection of 30 datasets (214,262 images) and used a different clustering algorithm for its splitting step, namely DBSCAN with a fixed maximal cluster size of 5. Between that preprint and the CVPR Workshop 2025 camera-ready version [2] we cite throughout this proposal, the authors revised the splitting algorithm to the single-linkage clustering method with a manually-set per-dataset threshold described below. We treat this revision itself as relevant evidence: the fact that the authors moved away from their original DBSCAN-based approach suggests they encountered limitations with it worth understanding, even though our own investigation targets the current, published algorithm.

13 of the 37 constituent datasets have timestamps at all,[2] a *similarity-aware split* is used everywhere else: images are embedded using DINOv2 [12], and for each identity, a graph is built where an edge connects two images if their cosine similarity exceeds a threshold θ; the connected components of this graph (found using a Union-Find/disjoint-set structure, i.e. single-linkage clustering) become the clusters, and whole clusters are assigned to the training set, while the remaining, unclustered images are split between training and testing.

The fact that images come from many different sources also affects re-identification directly: some animals are photographed in the wild, under uncontrolled ground and background conditions, while others are photographed in controlled environments, such as zoos or rehabilitation centres [3]. Because re-identification tends to be easier under controlled conditions, this difference in image source is itself a factor that any evaluation of the splitting methodology needs to account for.

In this proposal, our interest is specifically in this splitting methodology, and in particular in the similarity-aware split, since it is the one applied to the large majority of the dataset's constituent sources. We set out below the problems we see with the current approach, review the related work that informs our proposed investigation, and describe the approach we intend to take.

2 Problem Description

Our research is organised around four research questions, moving from whether the current similarity-aware split works, to whether it can be improved, to whether that improvement generalises and matters downstream:

RQ1. How effectively does the similarity-aware split prevent train-test leakage, compared with the time-aware split, on the datasets where both can be computed and compared directly?
RQ2. Can alternative threshold-selection and clustering strategies improve the reliability of the similarity-aware split relative to the current manually-tuned single-linkage method?
RQ3. Does a split methodology developed and validated on the 13 timestamped datasets remain reliable when deployed on the 24 datasets that have no timestamps to validate it against?
RQ4. How do improvements in split reliability, and training-time data augmentation applied within the resulting splits, affect downstream open-set and closed-set wildlife re-identification performance?

[2] Of these 13, the paper itself excludes 4 (AmvrakikosTurtles, ReunionTurtles, StripeSpotter, ZakynthosTurtles) from its own time-aware split comparison due to too few images or encounters, so the head-to-head evaluation we build on in P4 (Table 6 and Figure 5 of the paper) actually compares 9 timestamped datasets against the remaining 28 without timestamps, not 13 against 24. We use "13 timestamped datasets" loosely throughout this proposal to mean the set of datasets where timestamp metadata exists at all, and are precise about the smaller working set of 9 wherever we describe experiments that reproduce or extend the paper's own time-aware comparison.

These four questions correspond to four hypotheses we intend to test:

H1. The current similarity-aware split does not fully reproduce the leakage-correction effect that the time-aware split achieves on the same, timestamped datasets.
H2. Single-linkage clustering produces chained clusters containing image pairs whose direct pairwise similarity falls substantially below the clustering threshold θ.
H3. Automatic threshold selection and/or chain-resistant clustering methods produce higher cluster quality, measured against timestamp agreement, than the current manually-tuned single-linkage method.
H4. Split-quality improvements validated on the 13 timestamped datasets, together with training-only data augmentation, produce more reliable downstream Re-ID evaluation once deployed across the full, heterogeneous WildlifeReID-10k collection.

We break this down into seven concrete problems with the current methodology, grouped under the three limitations these hypotheses target: threshold selection (P1), clustering robustness (P2, P3), and validation/generalisation (P4, P5), plus two supporting concerns, imbalance (P6) and per-dataset heterogeneity (P7), that we treat as secondary to the central splitting problem.

Before setting out these seven problems, we want to be precise about what we mean by "leakage," since the term is used loosely in the paper and conflating its senses would weaken our own experimental claims. We distinguish three related but distinct notions: *image-level* leakage, where near-identical images (for instance, consecutive video frames) appear on both sides of the split; *encounter-level* leakage, where images from the same observation event appear on both sides even if no two images are near-duplicates; and *identity-level* leakage, where the same individual appears in both training and testing. These are not interchangeable. In closed-set re-identification, identity-level overlap between train and test is expected and necessary, since the task is precisely to recognise known individuals again; it is not leakage in the harmful sense. Image- and encounter-level leakage, however, can make the benchmark artificially easy regardless of the evaluation protocol, because the model may be relying on incidental cues (the same background, the same lighting, the same pose) rather than on identity-discriminative features. WildlifeReID-10k's splitting methodology is aimed squarely at removing image- and encounter-level leakage, and this is the sense in which we use the term throughout this proposal unless stated otherwise.

Relatedly, we adopt the following formal distinction between the two evaluation protocols the paper (and our proposal) reports results under. In the *closed-set* protocol, every identity that appears in the test set also appears in the training set, $ID_{\text{test}} \subseteq ID_{\text{train}}$, and the model's task is to identify the individual among a fixed set of known identities. In the *open-set* protocol, some test identities are deliberately withheld from training, $ID_{\text{test}} \not\subseteq ID_{\text{train}}$, and the model must additionally decide whether a query corresponds to a known identity at all, rather than being forced to pick the closest match among known individuals. Open-set evaluation is precisely where correct split construction matters

most, since if a withheld identity's images leak into training through an image- or encounter-level near-duplicate, the identity is no longer genuinely unseen, and the open-set protocol silently degrades back into a closed-set one for that individual.

P1: the similarity threshold is manually chosen, and this is only validated on a small, favourable subset of the data. WildlifeReID-10k's clustering step depends entirely on a cosine similarity threshold θ that decides which images count as near-duplicates. The paper reports that a single global threshold ($\theta = 0.97$) found 5,601 true-positive duplicate pairs against 685 false positives on the datasets with timestamps, but that manually choosing a different threshold for each dataset, based on visual inspection, improved this substantially, to 8,206 true positives against only 464 false positives, and this manual, per-dataset approach is what was actually used for the datasets without timestamps. This raises two separate concerns. First, "based on visual inspection" is not a documented, repeatable procedure; different people inspecting the same clusters could plausibly choose different thresholds, and the paper does not report what values were chosen for which datasets or how sensitive the results are to that choice. Second, and more importantly, this manual tuning is only checked against ground truth on the 13 datasets that have timestamps. For the remaining 24 datasets, which have no timestamps at all, there is no way, even in principle, to verify whether the manually-chosen threshold is any good, since there is no ground truth to compare it against. The manual method's advantage over the single global threshold is demonstrated only where it can be checked, and then applied on faith everywhere else.

P2: single-linkage clustering is structurally vulnerable to the chaining effect. The clustering step is not, in fact, DBSCAN, as an earlier preprint of this paper described; it is single-linkage clustering, implemented via connected components over a similarity graph. This is a different algorithm with a different, well-documented weakness. Single-linkage clustering builds clusters by transitively connecting any two points reachable through a chain of pairwise similarities above the threshold, so if image A is similar enough to image B, and image B is similar enough to image C, then A and C are placed in the same cluster even if A and C are not similar to each other at all. This is a long-recognised limitation of single-linkage clustering in the general clustering literature, commonly called the chaining effect [9]. In the context of WildlifeReID-10k, this matters because a long, gradually-changing sequence of images from a single encounter (an animal slowly turning, walking away, or moving in and out of shadow) is exactly the kind of chain that could merge visually very different images of the same individual into one oversized cluster, an outcome that could quietly inflate the size of the training-only cluster bucket well beyond genuine near-duplicates, without ever being flagged as a false positive under the paper's own quality metric, since that metric only checks whether a cluster's images share the same timestamp, not whether every pair of images inside the cluster is actually similar to every other pair.

P3: the cluster assignment is still a hard, binary decision. Despite the change in the underlying clustering algorithm, the fundamental character of the decision it produces has not changed: an image either belongs to a cluster or it does not, with no accompanying measure of confidence. The paper's own Table 3 shows that cluster purity (the share of a cluster's images sharing a single timestamp) degrades as clusters grow larger, from 97.5% for clusters of size 2 down to 77.4% for clusters of size 6 or more, which means that for the largest clusters, roughly one in five images placed inside them may not actually belong to the same encounter as the rest, yet every image inside the cluster is treated identically once it is assigned.

P4: the similarity-aware split recovers far less of the leakage-correction effect than the time-aware split does, on the paper's own evidence. This is, in our view, the most important open question raised by the paper's own results. The paper's Table 6 reports performance on the MiewID foundation model under four conditions: on the datasets that have timestamps, moving from a random split to a time-aware split closes a 12.3-point gap in closed-set accuracy (81.5% to 69.2%) and a 16.7-point gap in open-set performance (72.6% to 55.9%). But on the datasets without timestamps, moving from a random split to the similarity-aware split closes only a 1.5-point gap in closed-set accuracy (80.8% to 79.3%) and a 2.5-point gap in open-set performance (66.1% to 63.6%). The paper's own introduction explicitly frames the similarity-aware split as an extension of the time-aware split, meant to approximate its effect when timestamps are unavailable. If that is the intended relationship, this gap is difficult to reconcile: the similarity-aware split therefore appears to recover substantially less of the performance correction observed under the time-aware split. This could mean that the similarity-aware split is only catching a small fraction of genuine same-encounter leakage, or it could mean that the underlying datasets without timestamps are simply different in other ways, but the paper's experiments do not separate these two explanations, since split type and dataset group always co-vary in this comparison.

P5: cluster quality can only be checked where it is least needed. The paper's true-positive/ false-positive validation framework (Section 3.3 of the paper) can only be computed on datasets that have timestamps, because it needs a timestamp to check whether a cluster's images were really taken together. This means cluster quality is verified only on the 13 datasets that already have a better option available to them, the time-aware split, while the 24 datasets that actually depend on the similarity-aware split for their entire leakage protection have no way of confirming that their clusters are any good.

P6: the dataset is heavily imbalanced, and this is entangled with the wild-versus-controlled question. WildlifeReID-10k is heavily imbalanced across species and individuals, for example roughly 52,000 bird images spread across only 50 individuals against roughly 21,500 sea turtle images spread across 2,703 individuals, and this imbalance is not evenly distributed by capture condition either: about 50% of individuals in the dataset were observed in the wild (71% including rehabilitation centres), and re-identification tends to be noticeably easier in

controlled, good-lighting conditions than it is in the wild, which the paper's own per-dataset results reflect, since the best-performing datasets are mostly from controlled environments and the worst-performing ones are mostly from the wild. This raises a question we want to investigate directly: can data augmentation be used to help balance the dataset, so that species and individuals with very few images are not simply left underrepresented, and does augmentation narrow the gap between wild and controlled-environment performance, or does it mainly help the already better-performing controlled-environment datasets.

P7: aggregate metrics can hide how differently a method performs across the 37 source datasets. WildlifeReID-10k is a combination of 37 separate constituent datasets, each collected with a different camera, at a different image resolution, and under different environmental conditions, and each contributing very different numbers of images and individuals to the whole. The paper does report a per-dataset breakdown of BAKS for its baseline models (its Figure 4), which is a good existing practice, but this breakdown is not extended to the clustering step itself: we do not know whether the similarity-aware split's clusters are good on some source datasets and poor on others, only that the aggregate 76.1%-style purity figures (now the Table 3 purity breakdown) hold *averaged* over the 13 timestamped datasets. We are therefore interested in extending this per-dataset lens specifically to cluster quality, not just to downstream model performance.

3 Related Work

Self-supervised vision transformers (DINO and DINOv2). The clustering step used to build the similarity-aware split depends entirely on the quality of the image features it clusters, so the feature extractor itself is directly relevant related work, and it is important to keep two different papers distinct here. DINO [6] (2021) introduced a self-distillation approach for training vision transformers with no labels at all: a student network learns to match the representations a teacher network produces for a different augmented view of the same image, rather than being trained with human-provided labels; the teacher shares the student's architecture but is never updated by gradient descent, instead tracking an exponential moving average of the student's weights, and collapse (both networks trivially agreeing on a constant output) is avoided through centering and sharpening rather than any contrastive loss. This is self-distillation in DINO's specific sense, a student matching a teacher's representations under augmentation, and should not be confused with conventional knowledge distillation, which transfers a large model's predictions into a smaller model. The property of DINO that matters most for our proposal is that its self-supervised features cluster visually similar images together well, and its attention maps naturally highlight object regions without ever being told to. DINOv2 [12] (2023) is DINO's larger, later successor: it scales the same self-distillation idea to a bigger curated dataset and model size and produces stronger, more general-purpose visual features. It is specifically DINOv2, not the original DINO, whose embeddings WildlifeReID-10k uses to build its similarity graph, and DINOv2 features are also what the

paper uses directly as one of the tools to sanity-check the quality of the clustering itself.

Face re-identification (SurvFace). Cheng et al. [8] study a closely related re-identification problem in a different domain, introducing SurvFace, a benchmark for face re-identification built by mining face crops out of 17 existing person re-identification datasets rather than collecting new data. The paper draws a distinction that maps directly onto our own problem: standard face recognition models are trained and tested on clean, cooperative, high-resolution web photographs, while surveillance-style face re-identification must work with uncontrolled, low-resolution, poorly lit images, and models trained only on the former generalise surprisingly poorly to the latter. SurvFace also uses the same closed-set/open-set distinction that WildlifeReID-10k uses, and its metrics (FPIR, FNIR, TPIR@FPIR for open-set; CMC and rank-1 for closed-set) are not directly reused in WildlifeReID-10k. This is not because they would be technically inapplicable to animals, but because they assume a small watch-list style search and do not correct for severe class imbalance, whereas WildlifeReID-10k's BAKS/BAUS metrics were designed specifically to handle the extreme imbalance present in wildlife data. This is a useful, concrete illustration of a point that recurs throughout this proposal: the choice of evaluation metric should follow the shape of the underlying data.

Clustering-based pseudo-labelling and hard cluster assignment (Person Re-ID survey). Kumar et al. [10] survey supervised and unsupervised techniques for person re-identification, a problem that is structurally the same as animal re-identification: given a query image, retrieve the matching identity from a gallery of candidates. Two parts of this survey are particularly relevant to our proposal. First, unsupervised domain adaptation methods generate pseudo-labels through clustering algorithms such as DBSCAN or k-means, which confirms that clustering-based pseudo-labelling in general, not any one specific algorithm, is a standard tool across this wider literature. Second, and more directly relevant to WildlifeReID-10k's current algorithm, this survey identifies noisy pseudo-labels as a general weakness of hard, clustering-based assignment, arising both from domain gaps between source and target data and from the coarse, all-or-nothing nature of the assignment itself, regardless of which clustering algorithm produces it. Soft pseudo-labelling, where a probability of cluster membership is used instead of a binary in/out label, is proposed in this literature as the fix, and this remains a nuance that WildlifeReID-10k's current single-linkage clustering step does not explore, since connected components are, by construction, a hard partition.

Chaining effect and alternatives to single-linkage clustering. The specific weakness we raise in P2, that single-linkage clustering (which is what WildlifeReID-10k's connected-components procedure implements) is prone to transitively merging dissimilar points through an unbroken chain of intermediate similarities, is a long-established result in the clustering literature, going back to early critiques of the method and surveyed in general terms by Jain et al. [9]. A variety of fixes have been proposed for this specific weakness over the years, most of them

built around the idea of using local density to break spurious chains before they can form, filtering out low-density bridging points before running single-linkage, rather than using the raw similarity graph directly. This body of work is directly relevant to our proposed approach below, since density-adaptive clustering methods such as HDBSCAN [4, 5] were developed partly to address this exact failure mode in a principled way.

Automatic threshold and cluster-count selection. The manual, per-dataset threshold tuning we raise in P1 has a long-standing automatic alternative in the general clustering literature: the silhouette score [13], which measures, for each point, how much more similar it is to its own cluster than to the next closest cluster, and can be averaged across all points to score a candidate clustering without needing any ground-truth labels. Because it requires no labels, it is exactly the kind of automatic criterion that could replace visual inspection when choosing θ per dataset, and it is widely used elsewhere for this purpose in unsupervised settings where, as in WildlifeReID-10k's 24 datasets without timestamps, no ground truth is available to validate the choice directly.

Transformer-based re-identification and animal re-identification benchmarks. Ye et al. [15] provide a transformer-era follow-up to the Person Re-ID survey above, and its Animal Re-ID section is the most directly relevant part of the related work to our proposal. This section confirms that animal re-identification, as a field, is still immature compared to person or vehicle re-identification, with unclear task boundaries, species-specific methods that do not transfer well across species, and inconsistent experimental protocols across papers, which is essentially the same gap that motivated WildlifeReID-10k's own creation. In the same section, the authors also confirm that clustering-based pseudo-labelling remains the default approach for unsupervised re-identification even in transformer-based pipelines, which reinforces that the general weaknesses of hard, clustering-based assignment we describe above are a live, general methodological gap rather than something specific to wildlife data.

Open-set evaluation under class imbalance. Mendes Júnior et al. [11] propose an open-set classifier together with normalized accuracy and open-set F-measure, evaluation criteria designed to remain meaningful when the number of examples per class is highly uneven. BAKS and BAUS, the two metrics WildlifeReID-10k reports throughout, are themselves built on this line of work, which matters for our proposal because every one of our proposed experiments (P1–P7) is run over a collection whose constituent datasets vary by orders of magnitude in size and where naively-averaged accuracy would be dominated by whichever few datasets happen to be largest.

4 Proposed Approach

Our proposed approach is a single pipeline, applied identically to all 37 constituent datasets: images are optionally augmented, embedded with DINOv2, compared via cosine similarity, clustered, and split into training and testing sets. Timestamp availability plays no role in this pipeline itself; the same aug-

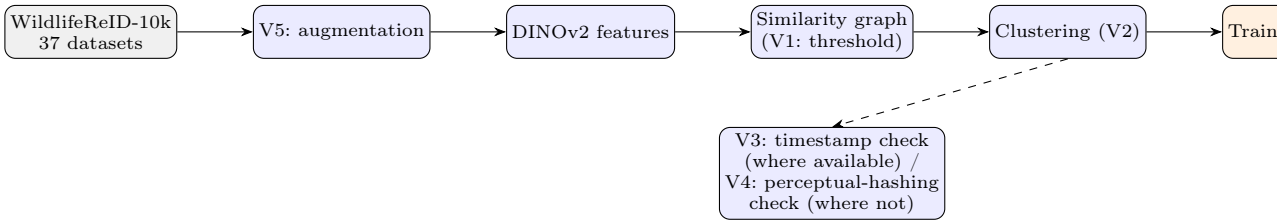


Fig. 1. Overview of the proposed pipeline. The same six steps, augmentation, DINOv2 feature extraction, similarity estimation, clustering, splitting, and evaluation, run identically for all 37 constituent datasets, regardless of whether timestamps are available. The dashed arrow shows validation applied afterward, as a check on the resulting clusters rather than a step in the pipeline itself: timestamp agreement where timestamps exist (V3), and perceptual hashing as an independent, model-free second opinion where they do not (V4).

mentation, feature extraction, similarity, and clustering steps run regardless of whether a given dataset happens to have timestamps. Timestamps are used only afterward, as a validation signal: for the subset of datasets where timestamps exist, we can check the resulting clusters against them to see whether the pipeline is behaving sensibly, in the same way the original paper does. This keeps the method itself general-purpose and deployable everywhere, while still letting us sanity-check it wherever a check happens to be possible.

Figure 1 summarises this pipeline. It is a single, linear sequence of six steps, augmentation, DINOv2 feature extraction, similarity comparison, clustering, splitting, and evaluation, run the same way for all 37 datasets. Timestamp checking is not one of these six steps. It is a separate validation check, shown below the main sequence in the figure, that we apply afterwards to see how well the clustering step worked, and only where timestamps happen to exist.

V1 Replace the manual, per-dataset threshold with an automatic one. For each dataset, we sweep candidate values of θ and score the resulting clustering with the silhouette score [13], which needs no ground-truth labels, and select the value that scores best. This directly targets P1 and H3: it removes the undocumented, visual-inspection step from threshold choice, and, unlike manual tuning, gives us a criterion we can also apply, and trust, on the 24 datasets that have no timestamps to check the threshold against.

V2 Directly test whether the chaining effect is a real problem in practice, not just a theoretical one, by running the current connected-components (i.e. single-linkage) clustering and a chain-resistant alternative, complete-linkage clustering, through the same pipeline on every dataset. Complete-linkage only merges two points into the same cluster if every pairwise similarity inside the resulting cluster stays at or above θ, which directly attacks chaining, since no chain of intermediate similarities can smuggle a dissimilar pair into a cluster, while remaining a standard, well-understood algorithm rather than one we would need to justify from scratch. Both variants are run identically

across all 37 datasets, using the threshold from V1; we then use whichever subset happens to have timestamps purely to check which variant agrees better with timestamp agreement, without that check being part of the pipeline itself. We will also report how often the current method's largest clusters (size 6+, where Table 3 shows purity already drops to 77.4%) contain image pairs whose direct pairwise similarity is far below θ, which would be direct evidence of chaining.

V3 On the subset of datasets that happen to have usable timestamps (9 of the 13 with timestamp metadata; see the footnote above), reproduce the comparison behind Table 6 as a validation check on the pipeline's output: random split, time-aware split, the current similarity-aware split, and our V2 alternative, all measured against each other on the same data. This directly answers whether an improved clustering step closes more of the 12.3/16.7-point gap that the current similarity-aware split only partially closes, without requiring the pipeline itself to treat timestamped and non-timestamped datasets any differently.

V4 The pipeline's own timestamp-agreement check (V3) can only be computed where timestamps exist. As a validation check on the pipeline's output for the datasets that lack them, we use an independent, model-free signal instead: perceptual hashing, a long-standing technique for near-duplicate detection unrelated to any learned embedding, and check where DINOv2-based clusters and perceptual-hashing-based near-duplicate flags agree or disagree. Disagreements are exactly the borderline cases worth flagging for extra scrutiny. We treat agreement between the two as an independent, model-free near-duplicate detection signal that increases our confidence in a cluster, not as proof that the cluster is correct: perceptual hashing can respond to shared background, composition, or pose rather than to the animal itself, and conversely the same individual photographed under very different conditions can hash very differently, so it is a useful second opinion rather than a ground truth in its own right.

V5 Investigate whether controlled data augmentation, applied *before* DINOv2 feature extraction and similarity analysis rather than only during Re-ID training, improves the robustness of the similarity representation used for clustering, with particular attention to underrepresented species and individuals (for example sea turtles relative to birds) and to wild-captured individuals relative to controlled-environment ones. Concretely, we generate augmented views (flips, crops, brightness/lighting changes, rotation) of each image, extract DINOv2 features for both the original and augmented views, and include the augmented views in the similarity graph and clustering step itself, rather than adding them only after the train/test split has already been fixed. This makes augmentation a controlled factor in the similarity-analysis pipeline: we compare a no-augmentation baseline against an augmented condition, with clustering, splitting, and downstream Re-ID training otherwise identical, and measure whether augmentation changes cluster quality, split quality, and the balanced-accuracy gap between well-represented and poorly-represented classes and between wild and controlled-environment datasets,

without simply inflating scores on the classes that are already well represented. We define a reproducible, non-arbitrary criterion for which individuals count as low-resource (fewer than a fixed threshold of training images) rather than hand-picking species after the fact.

As secondary, lower-priority directions that are already supported by the related work above but are not our primary focus, we note the possibility of a rotation-invariant backbone for the wild, uncontrolled-pose subset of the data, so that, for example, the same bear photographed once on land and once in a river is still recognised as the same individual regardless of background [14]; of wildlife-specific self-supervised pretraining as an alternative to using generic off-the-shelf DINOv2 features; and of dataset-aware clustering, which would treat each of the 37 constituent datasets as its own visual domain so that shared per-source style (lighting, background, camera setup) is not mistaken for true individual similarity.

Acknowledgments. This proposal is submitted as part of a group project undertaken at the University of Johannesburg.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this proposal.

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